

Machine Learning (IN2064)

Lecture 1: Introduction

Dr. Leo Schwinn (Data Analytics and Machine Learning)

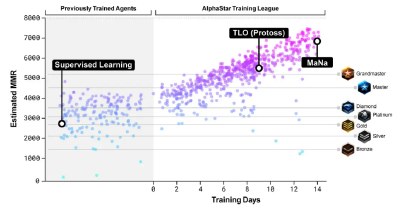
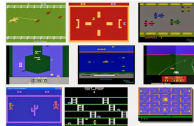
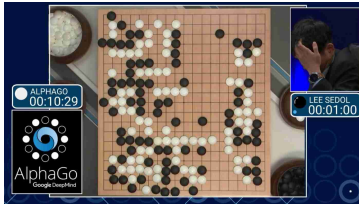
Prof. Dr. Stefan Leutenegger (Machine Learning for Robotics)

Technical University of Munich

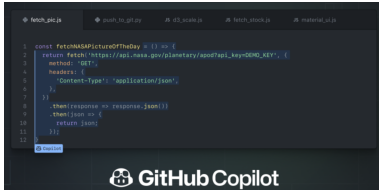
www.cs.cit.tum.de/daml/

Winter term 2024/2025

Game playing



Natural language processing



```
1 const fetchNASAPictureOfTheDay = () => {
2   return fetch('https://api.nasa.gov/planetary/apod?api_key=DEMO_KEY', {
3     method: 'GET',
4     headers: {
5       'Content-Type': 'application/json'
6     }
7   })
8   .then(response => response.json())
9   .then(json => {
10     return json
11   })
12 }
```

GitHub Copilot

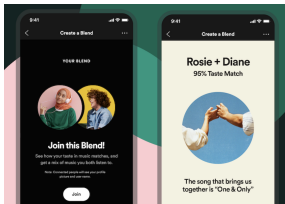


COPYWRITING SUMMARIZATION PARSING UNSTRUCTURED TEXT CLASSIFICATION TRANSLATION

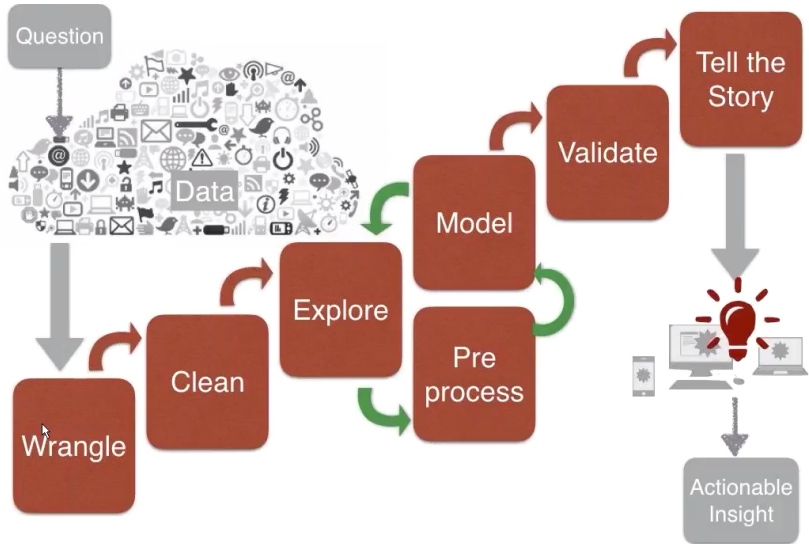
Create promo copy for the FamilyTime mobile application. It allows unlimited uploading, special filters and makes it easy to create albums of photos and videos. It runs on iOS and Android:

FamilyTime is the perfect app to share and create memories with your family. FamilyTime is a free, unlimited photo and video editor that allows you to create albums of your memories. It has a special filter for the perfect summer look and is so simple to use.

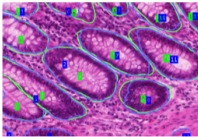
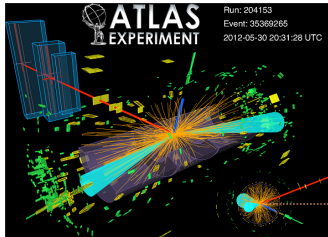
Web, ads and recommendations



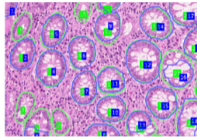
Data science



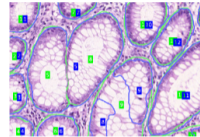
Physics, biology and medicine



(d) benign



(e) benign



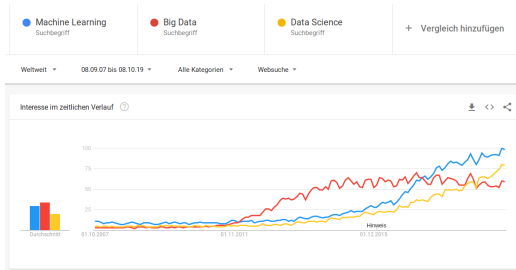
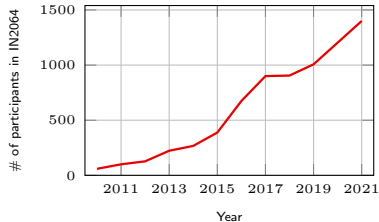
(f) malignant

What unites all these technologies?

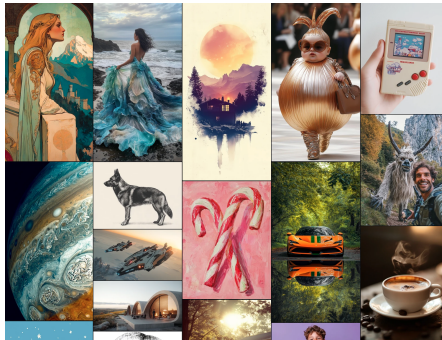
- Computer vision
- Natural language processing
- Recommender systems
- Computational advertising
- Robotics
- Artificial intelligence
- Data science
- Bioinformatics
- Many other fields

All are using Machine learning

Hot topic



Progress in the last 10 years



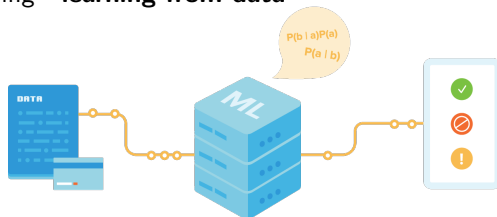
What is Machine Learning?

Simple example - classify transactions into **legitimate** and **fraudulent**.

Rule-based approaches - **rules** handcrafted by human experts

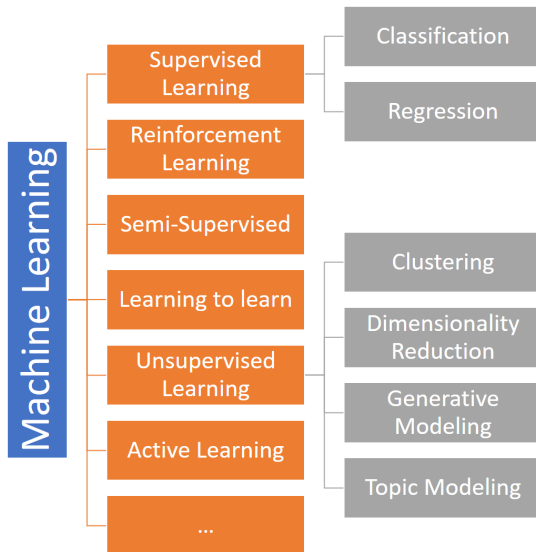


Machine learning - **learning from data**



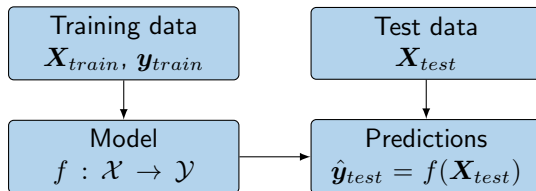
Figures adapted from <https://siftscience.com/sift-edu/prevent-fraud>

Types of ML problems



Supervised learning

- Given **training samples** $\mathbf{X}_{train} = \{\mathbf{x}_1, \dots, \mathbf{x}_N\} \subseteq \mathcal{X}$
- with corresponding **targets** $\mathbf{y}_{train} = \{y_1, \dots, y_N\} \subseteq \mathcal{Y}$
- Find a function f that generalizes this relationship, i.e. $f(\mathbf{x}_i) \approx y_i$.
- Using f , make predictions $\hat{\mathbf{y}}_{test}$ for the **test data** $\mathbf{X}_{test}$.



Supervised learning: Classification

If the targets y_i represent **categories**, the problem is called **classification**.

Examples

- Handwritten digit recognition
- Transaction classification
(**fraud**, **valid**)
- Object classification
(cat, dog, hotdog, ...)
- Cancer detection

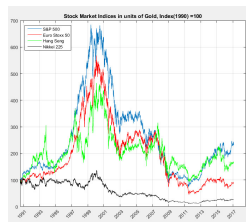


Supervised learning: Regression

If the targets y_i represent **continuous numbers**, the problem is called **regression**.

Examples

- Stock market prediction
- Demand forecasting
- User involvement measurement
- Revenue analysis

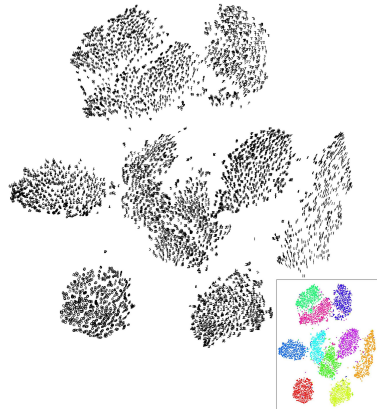


Unsupervised learning

Unsupervised learning is concerned with finding structure in **unlabeled** data.

Typical tasks

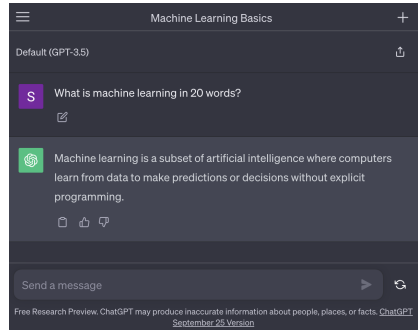
- Clustering
 - Group similar objects together
- Dimensionality reduction
 - Project down high-dimensional data



Unsupervised learning

Typical tasks - continued

- Generative modeling
 - (Controllably) generate new "realistic" data
- Topic models
 - Discover hidden semantic structures in text.



TEXT PROMPT an armchair in the shape of an avocado. . . .

AI-GENERATED
IMAGES



Edit prompt or view more images ↕

Other categories

- Reinforcement learning
 - Learning by interacting with a **dynamic environment**. Goal is to maximize **rewards** obtained by performing “desirable” actions.
- Semi-supervised learning
 - Learning to **combine lots of unlabeled data with a few labeled examples** for further prediction tasks.
- Active learning
 - Learn while obtaining labels by **querying an oracle**.
- Learning to learn (meta-learning)
 - Learning to **construct better models** for ML. Operates one level above the standard ML techniques.
- Learning to rank
 - What are the relevant items for a given query?
(e.g. Netflix, web search, ad placement)
- And many more...

General information

Staff

- Lecturer: Dr. Leo Schwinn & Prof. Dr. Stefan Leutenegger
- Teaching assistants:
Arthur Kosmala, David Lüdke, Marten Lienen, Tom Wollschläger,
Simon Geisler, Nicholas Gao
- Group website: www.cs.cit.tum.de/daml/

Details

- 8 ECTS, Language - English
- Open for both Bachelor and Master students
- Doesn't count for Wirtschaftsinformatik (Information Systems) students
- Doesn't stack with IN2332 in your curriculum

- Register at <https://piazza.com/tum.de/fall2024/in2064>
Access code: ml2024
- All announcements will be made on Piazza.
- All course material will be uploaded to Moodle.
- You will miss important information if you don't register.

Use Piazza to ask questions - your emails will likely not be answered.

Schedule & Logistics

- Lecture slides and exercise sheet for topic of week N are uploaded before Monday of week N .
- Monday & Tuesday: lecture for week N
- Wednesday morning: Q & A for week N
- Wednesday afternoon: exercise for week $N - 1$
- Exercise solutions are published on Moodle.

Exam

- Written final exam, probably in February
- Most likely in-person exam
- 120 minutes
- One handwritten two-sided A4 sheet with notes

Planned weekly schedule

Week	Date	Topic
1	Oct 16	Introduction & beginning of k-nearest neighbors
2	Oct 23	k-nearest neighbors, decision trees
3	Oct 30	Probabilistic inference
4	Nov 6	Linear regression
5	Nov 13	Linear classification
6	Nov 20	Optimization
7	Nov 27	Deep learning 1
8	Dec 4	Deep learning 2
9	Dec 11	Support Vector Machines
10	Dec 18	No Lecture
11	Jan 8	Dimensionality Reduction 1
12	Jan 15	Dimensionality Reduction 2
13	Jan 22	Clustering
14	Jan 29	Buffer

Contents

- This is an introductory, **theoretical** Machine Learning course
 - There will be a fair amount of theory and mathematics
 - We will focus on fundamental Machine Learning concepts
 - We will mostly discuss independent (iid) data
 - Next semester, we will cover (even) more advanced topics:
 - Machine Learning for Graphs and Sequential Data (IN2323)
 - Robustness
 - Sequential data
 - Graphs & networks
 - Advanced Machine Learning: Deep Generative Models (CIT4230003)
- These are the core research topics of our group :-)

What this course is not about

- This is **not** a pure Deep Learning course
 - look at IN2346 instead
- This is **not** a course about Big Data (Hadoop, etc.)
 - look at IN2326 instead
- This is **not** an applied Data Science / Business Analytics course
 - look at IN2028, IN2339 instead

Recommended reading

Our official reading recommendation:

- Christopher M. Bishop, *Pattern Recognition and Machine Learning*. Springer, Berlin, New York, 2006 (free, online version available).

but we also like:

- Kevin Murphy, *Machine Learning: A probabilistic perspective*. MIT Press, 2012.

What's next?

Brush up on your linear algebra, calculus, and probability theory knowledge.

Read

- <http://cs229.stanford.edu/section/cs229-linalg.pdf>
- <http://cs229.stanford.edu/summer2020/cs229-prob.pdf>
- Bishop [ch. 1.2.0 - 1.2.3, 2.1 - 2.3.0]
- Solve the math refresher (exercise sheet 1)